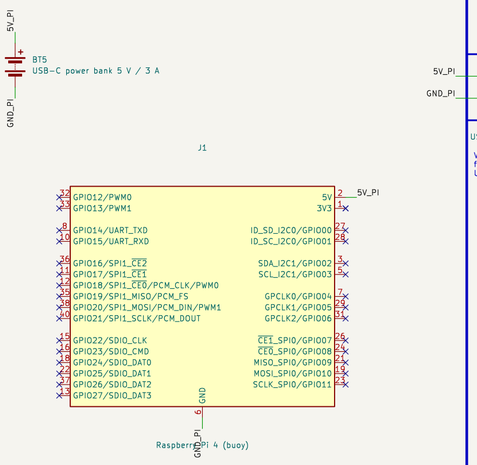
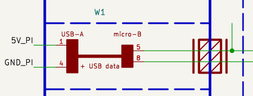
AA–TV wiring v11: single IP67 main box, 4x M16 cable glands, thruster with built–in ESC, 24 V bilge pump (water out) + solenoid valve (water in) on MOSFET switches, sealed ballast tank.

BUOY (surface, dry)



Pi 4 talks to the Pico over USB serial (/dev/ttyACM0) and powers it from USB 5 V. Reset the Pico by power-cycling the PI USB ports (uhubctl).

TETHER



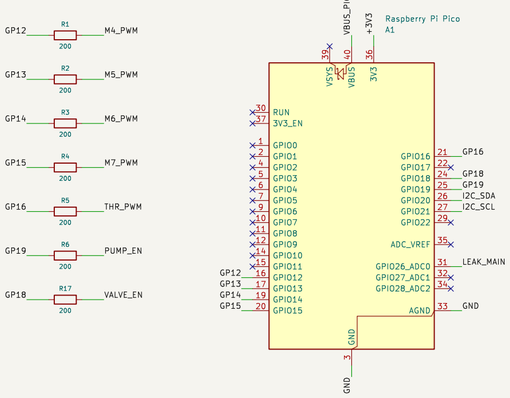
USB–A to micro–B, 3 m

Vinyl tube from the buoy is a sheath for the USB cable: it ends on G1. USB plugs are dry (buoy, Pico).

G1
G1: M16 gland, USB cable

MAIN BOX (IP67 ABS, 4x M16 glands; the only drybox)

BOARD: Pico + servo / signal solder–board



200 ohm in series on every signal that leaves the board (R1–R6, R17).



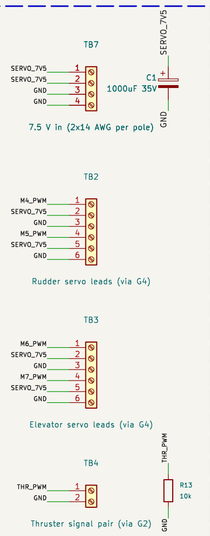
Leak sensor (box low point)

Balance lead in (JST–XH 4)

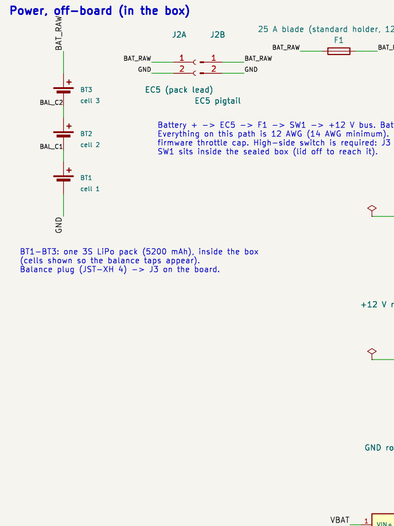
Board:
– One solderable breadboard. 7.5 V + GND rails: 14 AWG soldered along strips: servo + / – jumpers 14 AWG. Signal jumpers 20 AWG.
– No 12 V or 24 V on this board.
– R13: 10 k pull–down so the thruster ESC sees a clean low while the Pico boots.
– Pump and valve are plain on/off outputs (GP19, GP18). Never switch both on together.
– Free Pico pins: GP4, GP5, GP17, GP27, GP28.

ADS1115 16–bit ADC (breakout)

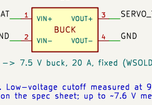
Battery monitor (balance lead –> J3):
– Dividers 100k/200k, 200k/100k, 100k/30k: 2.80 / 2.80 / 2.91 V at a full pack. Firmware scale x1.500, x3.000, x4.333.
– ADS1115 on I2C0 (GP20 SDA, GP21 SCL), addr 0x48, gain +/–4.096 V. The breakout has its own pull–ups and decoupling.
– Calibrate each channel once against a multimeter.
– C2–C4 (100 nF): fit when available. Unplug J3 for storage.



Power, off–board (in the box)

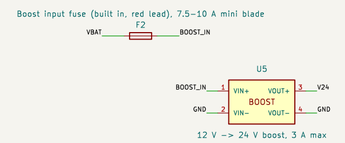


BT1–BT3: one 3S LiPo pack (5200 mAh), inside the box (cells shown so the balance taps appear). Balance plug (JST–XH 4) –> J3 on the board.



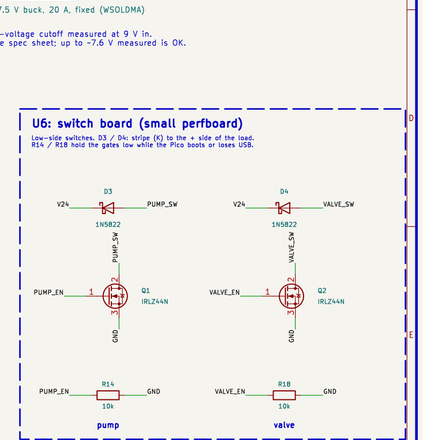
12/24 V –> 7.5 V buck, 20 A, fixed (WSOLDMA)

U1 output is fixed at 7.5 V. Low–voltage cutoff measured at 9 V in. Hitec DB777WP: 6.0–7.4 V on the spec sheet; up to ~7.6 V measured is OK.



12 V –> 24 V boost, 3 A max

24 V rail:
– U5 red/black = 12 V in, yellow/black = 24 V out. In – and out – must be common (check continuity).
– No fuse after U5: it limits its output to 3 A, which 20 AWG leads and the switches can carry.
– V24 feeds the pump + and the valve + directly. Q1 / Q2 switch the – side of each load.
– Valve drawn as 24 V. A 12 V valve must NOT go on V24: feed its + from the +12 V bus through a 2 A fuse.
– Optional: 1–2 A fuse in the valve + lead keeps the pump working if the valve shorts.



G4
G4: M16 gland, 4 servo leads (insert removed, epoxy–potted)

G2
G2: M16 gland, thruster cable + signal pair (potted)

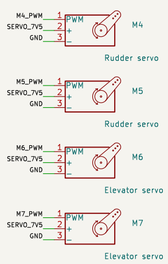
G3
G3: M16 gland, pump + valve leads (potted)

OUTSIDE THE BOX (wet): servos, thruster, bilge pump, solenoid valve

Gland rules (M16 x 1.5, clamp range about 4–8 mm):
– A gland seals ONE round cable. Anything else: pull the rubber insert, pass the leads, fill the gland body with epoxy (G2, G3, G4). Epoxy bonds poorly to silicone jackets.
– IP67 = 1 m for 30 min. Vacuum–test the closed box before every dive.

Ballast (sealed tank, no vent):
– Valve open: water in until the trapped air matches the outside pressure (about 9 % of the air space at 1 m). Open in short bursts.
– Pump on: water out. Valve closed + pump off + check valve: volume holds.
– Power or USB lost: pump off, valve closed.

Changed vs v9: thruster ESC (U2) is now inside the thruster. Pump ESC (U3). D2 Deutsch connector and the pump drybox are gone. Added boost U5, switch board U6 (Q1, Q2, D3, D4, R14, R18), valve SV1. R17, TB8. F1 30 A –> 25 A. Servo rail 7.4 V –> 7.5 V. Series resistors 220 –> 200 ohm. Dividers re–valued. R15 / R16 / C5 removed (on the ADS1115 breakout).

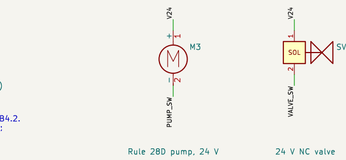


Servos: 2x Hitec DB777WP (4.5 A stall), 2x Savox SW–1250MG (2.4 A stall). All four stalled = 13.8 A on the 7.5 V rail. Both types tested on the Pico 3.3 V signal. Factory leads run uncut into the box through G4 and land on TB2 / TB3.



AplsQueen U92 Pro 12 V (ESC built in)

Thick pair: + to the +12 V bus, – to the GND bus (14 AWG). Thin pair: white = signal –> TB4.1, black = signal GND –> TB4.2. Cut off the XT60 to pass G2. 17 A at 12 V and 100 % throttle; cap the throttle in firmware. Max supply 16.8 V.



Rule 28D pump, 24 V

24 V NC valve

M3: bilge pump, 1.3 A at 24 V, one way: water OUT. Brown lead = +.
SV1: normally closed, direct–acting: water IN. Not chosen yet.
Plumbing: check valve on the pump outlet (water runs back through a stopped pump). Valve inlet and pump intake at the tank bottom.

High–side kill switch. F1 25 A, Servo rail 7.5 V, 24 V rail from the boost.
Glands: G1 USB, G2 thruster, G3 pump + valve, G4 servos (G2–G4 potted).
One IP67 box: battery, F1, SW1, bus bar, buck, boost, Pico board, switch board.

PIE AA–TV

Sheet: /
File: AATV_v11.kicad_sch

Title: AA–TV electrical v11: 24 V ballast, built–in–ESC thruster

Size: A2	Date: 2026–11–11	**Rev: v11**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 1/1